

ERIDSS Professional 5-Month Roadmap

Month 1 – Foundations

Weeks 1-2: Python OOP, JSON, File Handling, Git, GitHub.

Tutorials:

- Python Full Course (freeCodeCamp)
- Corey Schafer OOP Playlist
- Git & GitHub (Traversy Media)

Assignments:

1. Build CLI User Management System.
2. Parse JSON documents.
3. Push projects to GitHub.

Milestone:

Complete backend-ready Python skills.

Month 1 – Database & API Design

Week 3: PostgreSQL

Week 4: REST APIs, HTTP, API Design

Assignments:

1. Design ER Diagram.
2. Create Users, Documents, Entities, Relationships, RiskReports tables.
3. Write 30 SQL queries.

Milestone:

Finalize database schema.

Month 2 – Frontend Development

Weeks 5-8:

JavaScript ES6, React, Vite, Routing, Axios, Context API.

Assignments:

1. Login Page
2. Register Page
3. Dashboard
4. Upload Documents Page

Milestone:

Frontend UI completed.

Month 3 – Backend Development

Weeks 9-12:

FastAPI, Pydantic, SQLAlchemy, Alembic, JWT.

Assignments:

1. User APIs
2. Authentication APIs
3. Document APIs

4. Database migrations

Milestone:

Frontend connected to backend.

Month 4 – NLP & Knowledge Graph

Weeks 13-16:

spaCy, NLTK, NER, Dependency Parsing, NetworkX, Neo4j.

Assignments:

1. Extract entities from documents.
2. Generate relationships.
3. Build knowledge graph.
4. Store graph in Neo4j.

Milestone:

Risk graph generated automatically.

Month 5 – Analytics, Explainability & Deployment

Weeks 17-20:

Cytoscape.js, Graph Algorithms, Risk Scoring, Explainable AI, Testing, Docker.

Assignments:

1. Risk assessment engine.
2. Graph visualization.
3. Explainable recommendations.
4. Deploy with Docker.

Milestone:

Production-ready ERIDSS.

Recommended Folder Structure

```
backend/  
■■■ app/  
■ ■■■ api/  
■ ■■■ models/  
■ ■■■ schemas/  
■ ■■■ services/  
■ ■■■ auth/  
■ ■■■ database/
```

```
frontend/  
■■■ src/  
■ ■■■ pages/  
■ ■■■ components/  
■ ■■■ services/  
■ ■■■ hooks/  
■ ■■■ context/
```

```
nlp/  
graph/  
tests/  
docker/
```

Database Schema

```
Users(id, username, email, password_hash)  
Documents(id, title, filepath, uploaded_by)  
Entities(id, name, type)  
Relationships(id, source_entity, target_entity, relation_type)  
RiskReports(id, risk_score, explanation, document_id)
```

Project Deliverables

```
Phase 1: Authentication System  
Phase 2: Document Upload System  
Phase 3: NLP Processing  
Phase 4: Knowledge Graph  
Phase 5: Risk Intelligence Dashboard  
Phase 6: Explainable Decision Support  
Phase 7: Deployment & Documentation
```